THE ROLE OF EYE CONTACT IN HUMAN-ROBOT INTERACTION: TRUST AND DECISION-MAKING IN VIRTUAL FINANCIAL ADVISORY

Completed Research Paper

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Abstract

Eye contact is integral to social interactions among humans, signaling attention, building up trust, and ultimately driving human decision-making. As technology is advancing and robots are becoming more human-like, the question arises whether they can gain human trust and ultimately shape human decisions by human-like behavior such as making eye contact. This is especially crucial for the finance industry, where trust plays a major role and so-called robo-advisors establish themselves as a new tool for digital asset management. This research paper investigates the function of an anthropomorphic cue (eye gaze) in the context of a virtual financial advisory setting while following a 2 (averted vs. direct eye gaze) x 3 (human vs. robot-humanlike vs. robot-machinelike) experimental design (N=2,976). The results show that robots' eye gaze significantly increase peoples' perception of anthropomorphism, trust and satisfaction in virtual financial advisory. No impact was found on investment behavior.

Keywords: Eye Gaze, Robo-Advisor, Anthropomorphism, Investment Decision, Trust, Satisfaction.

1 Motivation

The integration of humanoid service robots into various industries is rapidly transforming the landscape of service provision (Belanche et al. 2019). With advancements in artificial intelligence, these robots are increasingly equipped with human-like skills, leading to their adoption across sectors. Notably, the financial advisory industry, renowned for its reliance on human expertise, is also witnessing the wide spread deployment of robo-advisors for digital asset management (Riya, Kanhaiya, and Vineet, 2023) and the first installation of humanoid robots as services providers in bank branches (Schaeder 2018).

As companies embrace humanoid service robots for their efficiency gains and cost reductions (Negahban and Smith, 2014), a pivotal concern arises: how will consumers respond to these automated service providers? Theoretically, humanoid service robots can offer many benefits to consumers such as convenience, flexibility, availability, and efficiency, but reduced trust may hinder consumer adoption (van Pinxteren et al., 2019). Trust forms the foundation of customer relationships, especially in the financial sector. Automating financial services while keeping customers' trust is challenging (Morana et al., 2020). Previous studies highlight the significance of trust in robo-advisors and its impact on consumer perceptions and financial decision-making (Jung et al. 2018, Hildebrand and Bergner 2021). To bolster trust, companies endeavor to enhance the human-likeness of robots (Waytz, Heafner, and Epley, 2014). Research indicates that perceived anthropomorphism, which augments trust, plays a pivotal role in shaping positive attitudes and purchase intentions towards digital assistants (Balakrishnan

& Dwivedi, 2021). Consequently, numerous humanoid robots have been developed with human-like facial expressions, voices, and names (Fink, 2012). However, the precise implications of such implementations remain ambiguous, as research on anthropomorphism reveals divergent outcomes across different contexts (van Pinxteren et al., 2019). We focus on eye contact as one of the most important social cues in human-human interactions (Kleinke 1986). Eye contact plays a vital role in establishing trust between individuals (Kaisler and Leder, 2016) and can impact decision-making processes, such as investment choices. Research reveals that individuals who establish direct eye contact with the target person are perceived as more trustworthy compared to those who do not make direct eye contact, highlighting the significance of eye contact in building trust and influencing behavior in online interactions (Broeder & Remers 2018). Given the observed enhancement of trust through direct eye contact in human interaction, we aim to explore whether this effect can be transferred to human-robot interaction Accordingly, we derive the following research question:

What impact does eye contact by an advisory robot have on financial decision-making?

To investigate this research question, this study employed an experimental, quantitative online approach. Participants engaged in simulated financial advisory conversations and then provided feedback. A research model predicting total and mediation effects was developed. The paper proceeds as follows: Section 2 reviews relevant literature. Section 3 presents hypotheses derived from literature and the research model. Section 4 explains the experiment, while Section 5 presents and discusses the results. Section 6 concludes with implications for research and management, along with limitations and future research directions.

2 Research Background

2.1 Humanoid robots

Humanoid service robots closely mimic the human form, possessing comparable physical structure, movement abilities, cognitive skills, and social interaction capabilities (Tuna & Ahmetoğlu, 2019). The Computers Are Social Actors paradigm suggests that individuals tend to anthropomorphize computational entities during human-computer interaction, attributing social traits to them (Nass, Steuer, & Tauber, 1994). Despite recognizing computers' lack of social needs, people persist in treating them as socially interactive (Reeves & Nass, 1996). Research indicates that humanoid robots are also subject to such anthropomorphism (Nass et al., 1995; Nass, Moon, & Green, 1995). However, while consumers generally favor robots with human-like attributes, there exists a limit to their acceptance, as proposed by the uncanny valley theory (Mori, 1970). This theory suggests that excessively human-like robots can evoke negative reactions, although its scientific validity is contentious.

Anthropomorphism plays a crucial role in shaping trust toward nonhuman entities (Luo et al., 2006). Trust can be described as an attitude wherein an agent is believed to aid in accomplishing an individual's objectives within situations marked by uncertainty, significantly influencing people's dependence on and decision to utilize technology (Lewis et al. 2018). As technology exhibits greater semblance to human mental states, individuals are inclined to trust it more in executing its designated tasks (Waytz, Heafner, & Epley, 2014). Several studies revealed that higher levels of anthropomorphic appearance (Kiesler et al., 2008, Luo et al., 2006), emotion (Brave, Nass and Hutchinson, 2005), memory (Richards and Bransky, 2014), and identity (Waytz, Heafner and Epley, 2014) lead to higher trust in robots. Similarly, research about recommendation agents has demonstrated that social presence enhances users' trust and even usage intentions (e.g., Hess, Fuller and Campbell, 2009; Lee and Choi, 2017; Qiu and Benbasat, 2009). For this purpose, many humanoid robots with human-like facial expressions, voices, and names have been developed (Fink, 2012). Although the impact of some human-like features on trust has been explored, it is unclear which features increase trust in which way. Feine et al. (2019) present a taxonomy of social cues for conversational agents. We focus on eye contact as one of the most important social cues in human-human interactions.

2.2 Eye gaze as an anthropomorphic feature

Eye contact plays a crucial role in social interaction between humans. It does not only signal turn taking in conversations but fulfills further functions such as expressing intimacy or exercising social control (Kleinke, 1986). It is a main nonverbal signal in social interaction. Special brain areas are dedicated to the interpretation of eye contact (Spezio et al. 2007). Eye contact also plays an important role in early childhood. Infants at 48h after birth prefer faces looking at them to faces looking away (Farroni, Menon, and Johnson 2007). Eye contact can be either unidirectional, where one individual looks at another without receiving a reciprocal gaze, or bidirectional which involves a mutual exchange of gazes between two individuals. When two people talk, they look at each other about 61% of the time, half of which even mutually (Argyle and Cook, 1976). Bidirectional eye contact especially signifies engagement and is critical for smooth interactions (Xu et al., 2016). Unidirectional eye contact has been found to affect attention and interest (Senju and Johnson 2009), memory (Mason, Hood and Macrae, 2004) and trust (Kaisler and Leder, 2016).

The deliberate use of unidirectional eye contact can be of great importance to marketers' intentions. In advertisements, for instance, eye contact by the spokesperson may increase product recall (Fullwood and Doherty-Sneddon, 2006), and the credibility of the ad (To and Patrick, 2021). As well as in sales, eye contact may increase consumer satisfaction (Andersson, Wästlund and Kristensson, 2016), buying intention (Broeder and Remers, 2018) and the judgement of the salesperson (Leigh and Summers, 2013). The well-explored domain of eye contact effects in human interaction has established a solid foundation for research. However, the question of whether these effects manifest similarly in human-robot interaction or entail perceptual distinctions represents a promising research area calling for further investigation.

In human-robot interactions, the dynamics of eye contact differ slightly. Studies have shown that human social brain activity is more pronounced during human-human eye contact compared to human-robot eye contact (Kelley et al., 2021). However, establishing eye contact is still considered a fundamental key to initiating interactions between humans and robots (Sharmin et al., 2021). Bidirectional eye contact with robots has been explored to make robots appear more engaging (Shimada et al., 2010) and likeable (Yonezawa et al., 2007) creating a better interaction experience (Kompatsiari et al., 2017). Unidirectional eye contact can influence users' arousal (Kiilavuori et al., 2021), attention and compliance (Admoni et al., 2014). Studying unidirectional eye contact in human-robot interaction offers a clearer and more controlled way to investigate the impact of the robot's gaze behavior on human participants without the interacting effects of reciprocal gaze exchanges. Against this background, we examine the yet unanswered question how unidirectional eye contact by a robot impacts consumers' trust and decision-making in service encounters.

2.3 Robo-advisors in finance

Robo-advisors in the finance sector are computer systems that offer investment recommendations based on individual financial information, without human intervention (Hodge, Mendoza and Sinha, 2021). These advisors gather client financial data through online surveys to propose and automatically execute suitable investments. Robo-advisors, serving as a tool for digital asset management, have seen significant growth. Their global market value, which was $4.51 billion in 2019, is expected to increase dramatically to $41.07 billion by 2027 (Riya, Kanhaiya, and Vineet, 2023).

Automating financial services by employing virtual agents or humanoid robots and maintaining customer trust is a great challenge (Morana et al., 2020). A recent study by Jung et al. (2018) demonstrated the importance of trust in automated financial services by robo-advisors. Participants requested a personal conversation with the human advisor before using the robo-advisor for their investment. In this conversation, participants did not ask for more information about the investment product but wanted to double-check the trustworthiness of the robo-advisor. The question arises if a human advisor is necessary to ensure the trustworthiness of the robo-advisor or if a more human-like robo-advisor might also increase the consumers' trust and invested amount.

The theory of Computers Are Social Actors (Nass, Steuer and Tauber, 1994) would suggest that increased human-likeness of the robot-advisor increases also consumers trust in the robo-advisor.

However, the study of Hodge, Mendoza and Sinha (2021) found that investors are more likely to adopt the recommendation of a robo-advisor with a low level of human-likeness. Considering the contracting evidence, the question arises how eye contact by a robot-advisor impacts consumers' trust and investment decision in financial services.

3 Hypothesis Development

3.1 Anthropomorphism

According to the Computers Are Social Actors theory, individuals tend to instinctively attribute social characteristics to computational entities during human-computer interaction (Nass, Steuer, & Tauber, 1994). Many humanoid robots have been developed with human-like appearance or behavior to increase human acceptance and trust (Fink, 2012). Our focus lies on eye contact as a primary nonverbal cue in both human-human interaction (Kleinke, 1986) and human-robot interaction (Sharmin et al., 2021).

Research consistently highlights the impact of eye contact on anthropomorphic perceptions. Studies show that in face-to-face interactions between robots and humans such as small talk or co-working, eye contact significantly increases anthropomorphic ratings compared to scenarios without eye contact (Zheng et al., 2015; Kuhnlenz, B., Wang, & Kuhnlenz, K., 2017; Kompatsiari et al., 2021, (Babel et al., 2021). This understanding underpins our first hypothesis.

H_1: Direct eye contact by a robo-advisor leads to significantly higher anthropomorphism than averted eye contact.

3.2 Trust

Trust plays a pivotal role in financial decision-making, particularly in contexts involving automated financial services such as virtual agents or humanoid robots (Morana et al., 2020; Jung et al., 2018). Human-like service robots are perceived not only as more friendly, intelligent, and likeable but also more trustworthy (Kalegina et al., 2019). Numerous studies have demonstrated that heightened levels of anthropomorphism correlate with increased trust in robots (Brave, Nass, & Hutchinson, 2005; Kiesler et al., 2008; Luo et al., 2006; Richards & Bransky, 2014; Waytz, Heafner, & Epley, 2014). Similarly, research about recommendation agents has demonstrated that social presence enhances users' trust (e.g., Hess, Fuller and Campbell, 2009; Lee and Choi, 2017; Qiu and Benbasat, 2009). Eye contact as one anthropomorphic cue has shown to impact trust levels, with people trusting robots more when they establish eye contact, especially in complex task scenarios (Stanton & Stevens, 2014) or when individuals are concerned about negative evaluation (Nomura & Kanda, 2015). This has also been found for virtual avatars in virtual environments (Normoyle et al., 2013). Based on these insights, the following hypothesis is proposed.

H_{2a}: Direct eye contact by a robo-advisor leads to significantly higher trust than averted eye contact.

Anthropomorphism also serves as a critical mechanism through which eye contact influences trust in human-robot interactions (Christoforakos et al., 2021). When individuals perceive robots as possessing human-like characteristics, such as the ability to make eye contact, they are more likely to anthropomorphize the robot, attributing human-like intentions, emotions, and capabilities to it which influences trust levels as individuals apply expectations governing human-human interactions to their interactions with robots (Pinxteren et al., 2019). This leads to the following hypothesis:

H_{2b}: The effect of direct eye contact by a robo-advisor on trust is mediated by anthropomorphism.

Establishing eye contact is fundamental to social interactions and triggers affective and attentional responses in both human-human and human-robot interactions (Kiilavuori et al., 2021). However, studies suggest differences in human reactions, with responses to human gaze being more pronounced compared to robot gaze (Kiilavuori et al., 2021). Additionally, in reflexive cueing tests, robots were found to evoke weaker responses compared to humans, indicating that cognitively, robots may be processed more similarly to inanimate objects like arrows rather than faces (Admoni and Scassellati,

2012). Based on these findings, it is anticipated that the effect of eye contact will be stronger in scenarios involving human interactions compared to those involving robots.
H_{2c}: Direct eye contact by a human-advisor has a stronger effect than direct eye contact of a robo-advisor on trust.

3.3 Overall satisfaction

Eye contact is recognized as a microskill that improves the effectiveness of advisory conversations (Barnett, Roach, and Smith, 2006). Positive outcomes, including enhanced satisfaction in advising contexts, have been linked to the use of these microskills (Daniels, 2003, Kaufman, 1975). In human-machine-interactions, higher level of perceived anthropomorphism significantly enhances positive attitudes (Balakrishnan and Dwivedi, 2021). Anthropomorphic social robots are seen as more collaborative and intelligent, making them preferred choices for assistance in various tasks (Kontogiorgos et al., 2019). In the context of financial advisory, it has been found that anthropomorphic robo-advisor do not only induce greater trust but also evoke a more benevolent evaluation of the service provider (Hildebrand and Bergner 2021). These findings inform the development of our hypotheses.
H_{3a}: Direct eye contact by a robo-advisor leads to significantly higher overall satisfaction than averted eye contact.
H_{3b}: The effect of direct eye contact by a robo-advisor on overall satisfaction is mediated by anthropomorphism.
H_{3c}: Direct eye contact by a human-advisor has a stronger effect than direct eye contact of a robo-advisor on overall satisfaction.

3.4 Investment amount

Financial decisions often trigger feelings of uncertainty and risk(Broeder and Remers, 2018). Establishing eye contact does not only foster trust (Kaisler and Leder, 2016), but may also affect decision-making. For instance, it can lead to higher chances of making a purchase (Broeder and Remers, 2018), increased expenditure (Powell, Roberts, and Nettle, 2012), and more daring behaviors (Wang and Lui, 2021). In human-computer interaction, the presence of trustworthy seller avatars maintaining eye contact has been shown to boost purchase behaviors among buyers (Bente et al., 2014). Furthermore, a heightened perception of anthropomorphism consistently amplifies intentions to purchase (Balakrishnan and Dwivedi, 2021). Similarly, social presence enhances users' trust and usage intentions (e.g., Hess, Fuller and Campbell, 2009; Lee and Choi, 2017; Qiu and Benbasat, 2009). In the realm of finance, anthropomorphic robo-advisors not only foster higher levels of trust and a more favourable perception of firms but also lead to greater acceptance of recommendations and increased investment amounts (Hildebrandt and Berger, 2021). Drawing from these findings, we formulate the hypothesis for the investment behaviors of participants in our study.
H_{4a}: Direct eye contact by a robo-advisor leads to significantly higher investment amount than averted eye contact.
H_{4b}: The effect of direct eye contact by a robo-advisor on investment amount is mediated by anthropomorphism.
H_{4c}: Direct eye contact by a human-advisor has a stronger effect than direct eye contact of a robo-advisor on investment amount.
In summary, our research embraces the total effects of eye gaze on trust, satisfaction, and investment amount with the corresponding mediation effects through anthropomorphism as well as the moderating role of the advisor type for the total effects (Figure 1). As our research model suggests, we don't investigate probable relations between three dependent variables describing consumer response but focus on the effects of an experimental manipulation of eye gaze.

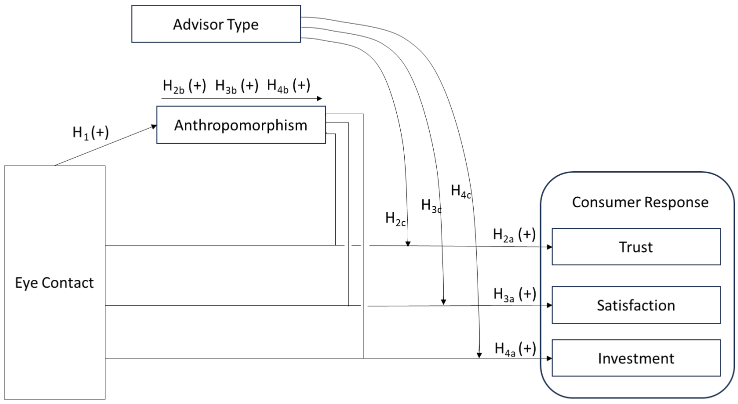

Figure 1. Research model.

4 Method

4.1 Setup

We follow a between-subjects design to analyze the impact of eye contact by a robot on consumer decision-making. We simulate an online financial advisory meeting in which participants receive investment advice from a financial advisor. Despite the advisor's gaze being transmitted through a 2D screen, mutual eye contact is assumed because this study fulfills the requirements of the Mona Lisa gaze effect. This effect states that as long as the projected head's gaze deviates by no more than approximately 5 degrees on either side, the gaze appears to follow an observer, irrespective of the observer's viewing angle (Moubayed, Edl and Beskow, 2012). For our research purpose, the experiment employed a 2 (eye gaze direction: direct vs. averted - left 28° and up slight 6°) x 3 (financial advisor: human agent vs. humanoid robot vs. machine-like robot) factorial design (see Figure 2). By comparing averted and direct eye gaze, we reveal how different forms of eye contact influence advisory meetings with robots, thereby striving to mitigate instances where the importance of advisors' gaze might inadvertently be overlooked, given the negative effects of averted eye contact in human interactions (Wirth et al. 2010). Due to the robot's restricted behavioral capabilities, we implement gaze aversion along with head aversion as standard practice in human-robot interaction (Kiilavuori et al., 2021), mirroring the simultaneous occurrence of these behaviors in human interactions (van der Wel 2018). The sample is divided into 6 groups, to which participants are randomly assigned before the start of the experiment.

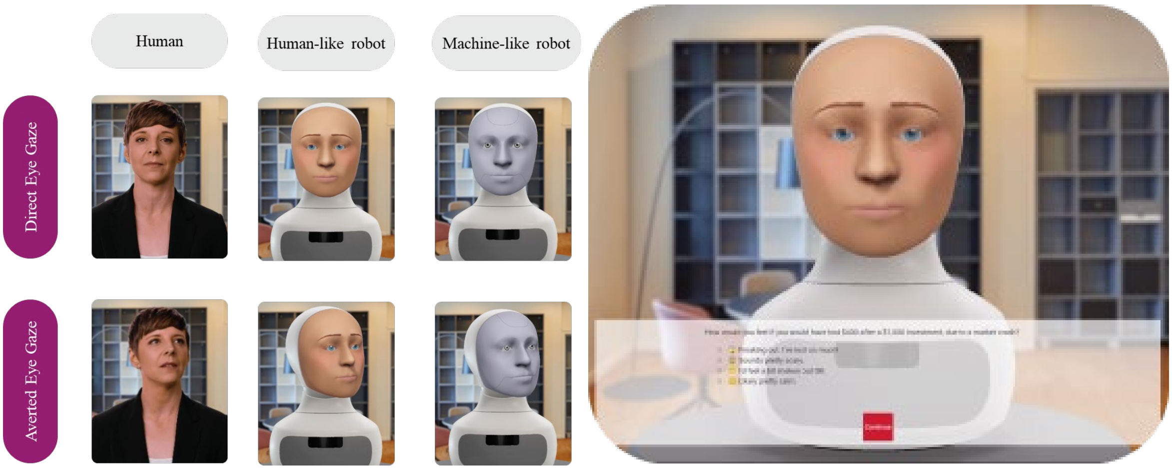

Figure 2. Experimental groups (left) and design of the user interface (right)

After an introduction to the study and a privacy notice, the experiment begins with the online advisory meeting. In the meeting, the advisor is generated via a seamless, dynamic sequence of pre-recorded short video segments consisting of speaking and idle segments of the different advisor types: the human advisor or, in other conditions, the Furhat Robot, whose appearance was customized as either human-like or machine-like. The user interface shows the advisor's video stream overlayed, when appropriate, with interactive user interface elements such as single-choice questions or number input fields, and non-interactive charts and figures used during the simulated consultation (see figure 2). In the meeting, only the advisor can speak. The participant interacts via clicks and enters monetary amounts. The financial advisor obtains information from the participants about their risk affinity, financial situation, and investment behavior. Subsequently, an investment plan is calculated and presented to the participant. To ensure comparability among the participants, this plan always aims at a fixed distribution of investment options (58% stocks, 33% bonds, 9% precious metals), whereby participants have the impression that they will receive an individualized plan based on their preferences. They then decide how much of the imaginary win (PLAY$ 10,000) to invest. After the investment decision is made, the next step of the experiment is to fill out a questionnaire about the experience of the online advisory meeting. The measures are presented in the follow-up section. To give the experiment a more realistic touch, each participant received a performance-based payout depending on the simulation of their investment based on real investment options from the past ten years (asset names and calculation formula are provided on request by the authors).

4.2 Measures

To validate the hypotheses, we applied quantitative methods to evaluate the experiment and the online survey. In the survey, we used different constructs to examine the individual perceptions of participants about the robo-advisor as well as the financial advisory meeting. The constructs in our questionnaire are based on previously validated items from existing literature. We customized these items to fit the specific context of our research, which revolves around the finance advisor. We measured the perception of eye gaze (Kiilavuori et al., 2021), anthropomorphism (Epley, Waytz and Cacioppo, 2007), trust (McKnight, Choudhury and Kacmar, 2002) and overall satisfaction (Fornell et al., 1996). We didn't evaluate the perception of anthropomorphism in the experimental group with the human advisor, because we expected respondents to be confused when answering this scale about the human. In addition, we included control questions to measure the perceived financial knowledge of participants (Gan, Khan and Liew, 2021) and whether the participants are familiar with the use of robots (Belanche, Casaló and Flavián, 2019) and have an affinity for technology interaction (Wessel, Attig and Franke, 2019). In the study, most constructs were measured using a 7-point Likert scale. The decision to use a 7-item scale, supported by several studies (Menold, 2018; Preston and Colman, 2000; Finn, 1972; Taherdoost, 2019), aimed to simplify participant responses and was deemed unlikely to cause fatigue due to the interactive nature of the experiment and financial incentives. The 7-point Likert scale response categories range from disagree to agree for each item used, except for the anthropomorphism scale, where the range was from not at all to very much. Moreover, we collected demographic data from the respondents to conduct a descriptive analysis of our experimental groups. An overview of the constructs including their respective measurement items are provided at the authors' website[1].

Construct		FL	CA	CR	AVE	SV
Perception of Eye Gaze	PEG	0.831 – 0.969	0.969	0.970	0.868	0.047 – 0.066
Anthropomorphism	A	0.719 – 0.948	0.946	0.949	0.789	0.062 – 0.197
Trust	T	0.724 – 0.871	0.952	0.954	0.655	0.066 – 0.631
Overall Satisfaction	OS	0.889 – 0.922	0.942	0.946	0.854	0.047 – 0.631

Table 1. Reliability and validity of constructs.

[1] Role_Eye_Contact_Measurement_Items.pdf (nim.org)

The constructs underwent a comprehensive evaluation of their reliability and validity (Table 1). We tested factor loadings (FL), internal consistency, and convergent validity. The measurement items loaded between 0.719 and 0.969 on their respective constructs, therefore exceeding the recommended threshold of 0.7 (Hair et al., 2018). The internal consistency of the scales showed Cronbach's alpha (CA) ranging from 0.942 to 0.969 and composite reliability (CR) ranging from 0.946 to 0.970, which exceeded the advised value of at least 0.6 (Hair et al., 2018), indicating a high internal consistency. The average variance extracted (AVE) ranged from 0.655 to 0.868, exceeding the recommended lower limit of 0.5, indicating convergent validity (Fornell and Larcker, 1981). To assess discriminant validity, we calculated for each factor the range of shared variances (SV) with other factors to ensure that the AVE of the factor is above this range (Fornell and Larcker, 1981).

4.3 Data collection

In June 2022, over two weeks, the online experiment was conducted. Participants for the study were acquired via research panel provider Prolific in the United States. They completed the experiment in about 15-20 minutes and received a fixed participation compensation of 2 GBP plus an additional amount of 1 to 1.53 GBP depending on their simulated investment. To determine the required sample size, we assumed an effect size of d = 0.20, alpha = 0.05, and power = 0.95 for our study, resulting in n=542 for each group (Zhang & Yuan, 2018). After removing the participants who failed to successfully complete the interaction session with the advisor, the final dataset consisted of a total of N=2,976, with a breakdown as shown in Table 2: 980 participants (32.93%) were distributed to the human-advisor group, 1,010 participants (33.94%) to the robo-advisor that looked humanlike, and 986 participants (33.13%) to the robo-advisor that looked machinelike. The average age of the participants was 37 years, with a standard deviation of 13. Regarding gender distribution, 1,657 (55.68%) of the participants identified as male, while 1,319 (44.32%) identified as female. Most of the sample completed a bachelor's degree (N=1,197, 40.22%) as their highest educational attainment.

Advisor	**Human-Advisor**		**Robo-Advisor: human-face**		**Robo-Advisor: machine-face**		**balance test: direct vs. averted, p-value**[5]
Gaze	**Direct**	**Averted**	**Direct**	**Averted**	**Direct**	**Averted**	
N[1]	488	492	518	492	495	491	
Gender[2]	245|237|10	240|235|13	327|183|8	282|205|5	255|228|12	249|231|11	0.173
Age[3]	36.88 (13.84)	36.35 (12.86)	37.53 (12.64)	38.36 (13.06)	35.07 (12.7)	35.06 (12.21)	0.828
Education[4]	0.60 (0.49)	0.64 (0.48)	0.61 (0.49)	0.62 (0.49)	0.61 (0.49)	0.60 (0.49)	0.601
PEG[3]	5.99 (1.18)	2.18 (1.468)	5.20 (1.46)	2.39 (1.43)	5.01 (1.59)	2.41 (1.48)	<0.001

Note: [1]per group, [2]male | female | other, [3]Mean value (Standard deviation), [4]share academic degree, [5]F-test in two-way ANOVA Advisor-Type x Gaze

Table 2. Descriptive statistics of collected data.

5 Results

5.1 Randomization and manipulation check

The randomization of gaze manipulation assignment was effective. Based on the results of the F-test in a two way ANOVA by advisor type and gaze condition, we found overall no significant differences (at 5% significance level) between experimental gaze conditions in socio demographic variables (Table 2) and in potential confounders capturing perceived financial knowledge (p = 0 .721), affinity for technology interaction (p = 0.241), familiarity with the use of robots (p = 0.742), previous experience with human (p = 0.403) or robotic financial advisors (p = 0.461). Furthermore, there were no significant differences for these variables between experimental gaze conditions within each advisor type, indicating effectiveness of randomization.

The experimental manipulation of direct vs. averted gaze was effective overall (Table 2) as well as for each advisor type as indicated by significant differences (by Welch's t-test) between experimental groups in Perception of Eye Gaze (PEG): Human-Advisor ($p < 0.001$), Robo-Advisor with human-face ($p < 0.001$) and Robo-Advisor with machine-face ($p < 0.001$).

5.2 Hypothesis testing

A two-way ANOVA by robot type and gaze, including the interaction, indicated a statistically significant effect (at 5% significance level) of directed vs. averted gaze on anthropomorphism perception for robot advisors ($p<0.001$, Cohen's-f = 0.08, CIf95% = [0.05, Inf]). The not significant interaction term ($p=0.549$) and the follow-up contrasts (by Welch's t-test) with Holm correction of the p-values for family-wise test errors (Table 3) imply that this effect is comparable and positive for both robots. In summary, the results provide supporting evidence for hypothesis H_1 for both robots.

	M_{direct}	$M_{averted}$	p-value (one-sided)	Cohen's-d with 95% CI
Machine-face	2.44	2.24	0.011	0.15 [0.04, Inf]
Human-face	2.58	2.30	0.004	0.18 [0.08, Inf]

Table 3. Anthropomorphism perception for directed vs. averted gaze.

The manipulation of gaze had a statistically significant effect on perceptions of trust across the advisors as indicated by the two-way ANOVA with interaction ($p<0.001$, Cohen's-f = 0.09, CIf95% = [0.06, Inf]). The Holm-corrected p-values of the follow-up contrasts suggest that this effect was significant and positive for both robots (Table 4). Although the effect seems to be slightly stronger for the human advisor than for the human-face robot, and particularly for the machine-face robot, the interaction term in the ANOVA was not significant ($p =0.416$).

	M_{direct}	$M_{averted}$	p-value (one-sided)	Cohen's-d with 95% CI
Human	5.52	5.23	<0.001	0.24 [0.14, Inf]
Machine-face	5.04	4.90	0.045	0.11 [0.00, Inf]
Human-face	5.24	5.01	0.005	0.18 [0.07, Inf]

Table 4. Trust perception for directed vs. averted gaze.

Thus, the results provide only partial support for H_{2a} hypothesis. Particularly, the statistical analysis of our data supports the positive effect of direct eye contact for the human-face robot, whereas the results for the machine-face robot are mixed, showing the 95%-confidence interval for the effect size including 0 (and not significant total effect in mediation analysis as will be shown in Table 6 below).

In a separate set of follow-up contrasts for interaction term, we examined the differences in the effect of gaze between the experimental groups with human vs. robotic advisor. For this purpose, we tested the hypotheses of no differences in gaze effects against one-sided alternative hypotheses of the form:

$$M_{direct}^{Human} - M_{averted}^{Human} > M_{direct}^{Robot} - M_{averted}^{Robot}$$

For these contrasts, no correction of (one-sided) p-values for family-wise errors is applied for the Welch's t-tests to be more sensitive in detecting of potential differences. The results showed no significant difference (at 5% significance level) in the gaze effects between the human advisor and the robot with machine-face ($p = 0.094$), and no significant difference between the human advisor and the robot with human-face ($p = 0.293$). These results don't support hypothesis H_{2c}, suggesting that the gaze effects on trust perceptions are rather comparable for human and robotic advisors.

To examine the hypothesis H_{2b}, that the effect of gaze on trust occurs due to a more anthropomorphic perception of a robotic advisor, we conducted a causal mediation analysis (Imai et al., 2010). We estimated a single mediator model for each advisor type with gaze as treatment, anthropomorphism perception as mediator and trust perception as outcome using R-mediation software package (Tingley et al., 2014). For estimation of standard errors and confidence intervals of the effect estimates we used non-parametric bootstrap (with 5000 draws) (Hayes et al., 2008; VanderWeele, 2015). The causal interpretation of the estimates is based on the assumption of no unmeasured or treatment-induced mediator-outcome confounders (VanderWeele, 2015), for which we found no theoretical or empirical objections for the corresponding variables. The resulted regression estimates in Table 5 indicate that the direct effects for both robots are not significant in contrast to the mediated effects. Thus, the effect of gaze on trust were significantly mediated by anthropomorphism perception as supposed by the hypothesis H_{2b}.

	Total effect		Natural direct effect		Natural indirect effect	
	β with 95% CI	p-value	β with 95% CI	p-value	β with 95% CI	p-value
Machine-face	0.14 [-0.03, 0.30]	0.091	0.06 [-0.09, 0.20]	0.410	0.08 [0.01, 0.15]	0.020
Human-face	0.25 [0.06, 0.38]	0.005	0.13 [-0.01, 0.27]	0.076	0.10 [0.03, 0.17]	0.004

Table 5. Mediation analysis of gaze effect on trust through anthropomorphism perception.

Similar to the perceptions of trust, we found a statistically significant effect of direct vs. averted gaze on the overall satisfaction with the financial advisory system across all advisor types ($p<0.001$, Cohen's-f = 0.09, $CI_f^{95\%}$ = [0.06, Inf]). The effects were significantly positive for each advisor as shown by the follow-up contrasts with Holm correction of p-values and the confidence intervals of these effects in Table 6. Thus, the results support our hypothesis H_{3a}.

	$\mathbf{M_{direct}}$	$\mathbf{M_{averted}}$	**p-value (one-sided)**	**Cohen's-d with 95% CI**
Human	5.67	5.45	0.016	0.15 [0.05, Inf]
Machine-face	5.51	5.22	0.002	0.21 [0.10, Inf]
Human-face	5.53	5.33	0.016	0.14 [0.04, Inf]

Table 6. Overall satisfaction for directed vs. averted gaze.

To test the hypothesis H_{3c}, that direct eye contact of a human advisor has a greater effect than direct eye contact of a robo-advisor, we tested (similiar to the analysis of trust) a separate set of follow-up contrasts for differences in gaze effect between different advisor types. We found no significant difference between the gaze effects for the human advisor and for the robot with machine-face ($p = 0.725$) as well as no significant difference between the human advisor and the robot with human-face ($p = 0.433$). In summary, the data provide no support for our hypothesis H_{3c}.
Like the investigation of the effects on trust, we conducted a causal mediation analysis using non-parametric bootstrap (5000 draws) to determine the extent to which the gaze effect on overall satisfaction is mediated by anthropomorphism perception (see Table 7).

	Total effect		Natural direct effect		Natural indirect effect	
	β with 95% CI	p-value	β with 95% CI	p-value	β with 95% CI	p-value
Machine-face	0.30 [0.11, 0.48]	<0.001	0.23 [0.05, 0.39]	0.011	0.07 [0.01, 0.13]	0.020
Human-face	0.20 [0.02, 0.37]	0.030	0.12 [-0.05, 0.28]	0.170	0.08 [0.03, 0.14]	0.004

Table 7. Mediation analysis of gaze effect on Satisfaction through anthropomorphism perception.

The results in Table 7 support our hypothesis H_{3b}, that the gaze effects are significantly mediated by anthropomorphism for both types of robo-advisors. Additionally, the estimates revealed a significant and stronger direct, i.e., not mediated, effect for the robot with machine-face, thus, raising the question of underlying psychological mechanism of this effect.

A two-way ANOVA by robot type and gaze including interaction indicated no statistically significant effect of directed vs. averted gaze on investment amount across the advisors (p=0.914, Cohen's-f = 0.00, $CI_f^{95\%}$ = [0.00, Inf]). Also, the follow-up contrasts for each advisor type revealed no significant effects (Table 8). Thus, we had no evidence in support of the hypothesis H_{4a}.

	M_{direct}	$M_{averted}$	p-value (one-sided)	Cohen's-d with 95% CI
Human	6379.9	6518.2	0.758	-0.04 [-0.15, Inf]
Machine-face	6360.3	6267.5	0.316	0.02 [-0.07, Inf]
Human-face	6500.7	6496.5	0.492	0.00 [-0.10, Inf]

Table 8. Investment amount for directed vs. averted gaze.

In view of the absence of significant effects for any advisor type we skipped examination of the hypotheses H4b and H4c, and consider them as not supported by our data. Since we observed a significant positive effect of gaze on trust, and trust is generally considered an important predictor of investment decisions, it is surprising that there is no evidence for the gaze effect on investment amount. We attribute this finding to the high variation in the investment amount variable and to its weak overall association with the trust variable (r = 0.039) in our data.

6 Conclusion and Discussion

6.1 Summary

Our study examined the eye gaze of humanoid robo-advisors as an anthropomorphic cue within an online advisory setting. The 2x3 online experiment included 2,976 participants who took part in a financial advisory meeting and filled out a follow-up questionnaire. Besides theoretical concepts, we additionally investigated behavioral measures by simulating an actual investment decision. Our results show that the impact of direct eye contact (vs. averted gaze) is significant for trust in human-robot interaction, especially for more human-like robots. In addition, participants reported significantly higher overall satisfaction after the advisory meeting with the financial robo-advisor who looked them in the eye. Interestingly, the effect of direct eye contact by a robo-advisor is comparable with the effect of direct eye contact by a human advisor regarding both trust and satisfaction. A statistically relevant difference between the experimental groups regarding their investment amount cannot be observed. The additional investigation of the robo-advisor's level of anthropomorphism led to further findings. We found that the relationship between eye gaze and trust as well as eye gaze and satisfaction is significantly mediated by the anthropomorphism of the robo-advisor.

6.2 Theoretical contribution

Our findings align with previous research that emphasizes the importance of anthropomorphism in human-robot interaction. Studies have shown that factors such as robot competence and warmth positively influence trust development in robots (Christoforakos et al., 2021). Additionally, the level of anthropomorphism of robots has been found to impact consumers' perceptions and acceptance of service robots (So, 2023). Despite this, the impact of anthropomorphism on human behavior remains a topic of ongoing debate and investigation. While some studies report significant effects of anthropomorphism on behavior (Hildebrandt and Bergner 2021), others offer only partial support (Martelaro et al., 2016), and some find no effects at all (Salem et al., 2015). In contrast to human attitudes, which tend to be more stable and consistent, human behavior is susceptible to various situational influences and contextual factors (Salem et al., 2015). This variability is particularly evident in financial advisory contexts involving robots, where previous experience is limited, and uncertainty looms large (Morana et al., 2020). These findings underscore the need for further research to elucidate the complex relationship between anthropomorphism, trust, and behavior in human-robot interaction (Lewis et al., 2018). In conclusion, the research results provide valuable insights into the role of direct eye contact and anthropomorphism in human-robot interaction. By demonstrating the significance of these factors in shaping trust and satisfaction levels, the study contributes to a deeper theoretical understanding of how design elements and anthropomorphic features influence human perceptions and interactions with robots, particularly in the context of financial advisory, and calls for more research on the impact of anthropomorphism on human behavior.

6.3 Practical implications

The potential of service robots to revolutionize the service industry is considerable, with their ability to operate round-the-clock and deliver efficient service. However, the critical question is whether consumers will readily accept them (van Pinxteren et al. 2019). Our study highlights the significance of eye contact in fostering trust and satisfaction, not only in human interactions but in robotic interactions as well. Service providers employing robotic advisors should integrate eye contact and other non-verbal cues, such as nodding and smiling at appropriate moments. These features should be as realistic as possible to enhance the overall consumer experience and trust in robotic services (Andrist, Mutlu and Gleichner, 2013).
Consumers should recognize their inclination to trust AI that acts more human-like. Understanding that even subtle social cues, like a robot making eye contact, can influence attitudes is crucial. It's important for consumers to be conscious of these effects. Being informed about these developments will aid in making educated choices about utilizing these services. For society as a whole, comprehending AI's role in influencing consumer choices is crucial for shaping discourse and guiding potential regulations in this emerging area.

6.4 Limitations and future research

This study, while shedding light on the role of eye gaze in humanoid robo-advisors, is subject to certain limitations that pave the way for future research. Firstly, the experimental design was confined to the use of only two robots (Furhat models) and a single human. This limited range may not fully represent the wide spectrum of possible interactions with various robotic and human characters. Additionally, the investment scenario presented to participants was simulated, not real, potentially influencing the authenticity of their responses and decisions. A significant cultural consideration arises from the fact that our findings are based on data from the United States. Eye contact has varied interpretations and significance across different cultures. In human conversations, people from Arab, Latin American, and Southern European have found to engage more often in eye contact than people from Asia or Northern Europe (Argyle and Cook, 1976). This might impact the generalizability of our results. In Japan, where making eye contact is traditionally seen as impolite (Nixon and West 1995), the use of eye contact with a robot might not foster trust among Japanese consumers. This cultural aspect underscores the importance of considering regional variations in non-verbal communication norms in designing humanoid robots.
As robotic technology is still in its nascent stages, public familiarity and comfort with robot interactions are evolving. Future research should thus consider the changing dynamics of human-robot relationships as societal adoption grows. Moreover, our study focused specifically on a financial advisory context, raising questions about the applicability of our findings to other service settings such as shopping malls or restaurants. It remains to be explored whether the effects of eye contact by robo-advisors would be consistent across different settings. The influence of eye contact might vary depending on the context, as research in anthropomorphism has revealed different effects in different situations (van Pinxteren et al., 2019).
Addressing these gaps, future studies should aim to diversify the types of robots and human representations used, incorporate real-life investment scenarios, examine the cultural nuances in eye contact perception and explore the impact of robotic eye contact in various situational contexts as we advance towards a more robot-integrated society. Understanding these aspects is vital for enhancing the efficacy and acceptance of robotic advisors in diverse sectors.

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